

Supplementary Material for "Unobtrusive Web-Based Sensing Framework for Longitudinal Affective Monitoring"

I. FACEMESH LANDMARK INDICES

The MediaPipe FaceMesh model returns 468 landmarks. For calculating the metrics used in this study, we utilized the specific indices listed in Table ???. The "Rigid Alignment" indices were selected based on their low variance during facial expression (i.e., they move with the head, not the skin).

TABLE I
SELECTED FACIAL LANDMARK INDICES

Region of Interest	Mesh Indices (MediaPipe)
Nose Tip (tracking)	1
Left Tragus (rigid)	234, 127, 162
Right Tragus (rigid)	454, 356, 389
Left Eye (EAR)	33, 160, 158, 133, 153, 144
Right Eye (EAR)	362, 385, 387, 263, 373, 380
Forehead (rPPG ROI)	10, 338, 297, 332, 284, 251, 389, 356

II. SIGNAL PROCESSING SPECIFICATIONS

The raw physiological signals extracted from the vision pipeline are inherently noisy. We applied the following digital signal processing (DSP) chain using the WebAudio API's 'AudioContext' for real-time filtering, or 'scipy.signal' for post-processing.

A. rPPG Bandpass Filter

To isolate the cardiac frequency band from motion artifacts and DC offsets:

- **Type:** 4th-order Butterworth Filter (Infinite Impulse Response).
- **High-Pass Cutoff:** 0.7 Hz (42 BPM). Removes slow drifts due to shadow movement.
- **Low-Pass Cutoff:** 3.0 Hz (180 BPM). Removes 60Hz power line noise and sensor thermal noise.
- **Zero-Phase Implementation:** Applied bidirectionally ('filtfilt') to prevent phase distortion which would alter HRV timing.

B. Motion Smoothing

The raw head coordinate P_t was smoothed using an Exponential Moving Average (EMA) to reduce quantization noise from the webcam sensor:

$$S_t = \alpha \cdot P_t + (1 - \alpha) \cdot S_{t-1} \quad (1)$$

Where $\alpha = 0.2$, equivalent to a time constant of $\tau \approx 80ms$ at 60 FPS. This preserves the sharp onset of a "Freeze" response while smoothing out jitter.

III. SYSTEM LATENCY ANALYSIS

A critical requirement for psychophysics is timing precision. We profiled the "End-to-End" latency of the Camerale system using a high-speed camera (240 FPS) observing the screen-to-photodiode transition.

- 1) **Input Latency:** $8.4ms \pm 2.1$ (USB Keyboard via 'key-down' event).
- 2) **Display Latency:** $16.7ms$ (Fixed by 60Hz V-Sync).
- 3) **Processing Latency:** $12.3ms \pm 4.5$ (MediaPipe Inference Time on Intel Iris Plus Graphics).

Total System Latency is approximately $37.4ms$. Since the Reaction Time (RT) analysis searches for differences $> 50ms$ between conditions, this latency floor is acceptable.